will be public, and many will be a combination of the two. This will lead to a highly transient experience, both as consumers and increasingly as workers. And, therefore, SSI is such an exciting and important facilitation component for the metaverse.

SSI will handle standard authentication requirements, but so much besides. It has the potential to support the concept of 'living resumes,' which will accurately and exhaustively capture your work, education, and deliverable history. Background verification checks would be made near instantaneous, as all this information could be kept up-to-date and validated through SSI.

Another potential application is pensions, pension contributions, and employment benefits, something that will prove increasingly difficult to track as employment tenures continue to shorten.

There is an incredibly significant level of contribution being made by the open source community to SSI. Notable examples are:

- The European Self Sovereign Identity Framework Lab (eSSIF-Lab)
- Hyperledger Foundation (under the Linux Foundation)
- Trust Over IP Foundation (under the Linux Foundation)
- IDUnion
- Sovrin Foundation
- World Wide Web Consortium (W3C)
- Decentralised Identity Foundation (DIF)
- Network of Networks

Smart contracts. Another fundamental application of blockchain technologies that will be a critical component of the metaverse is smart contracts.

There are numerous potential applications of smart contracts that will help resolve usual challenges and areas of contractual inefficiency—for example, statements of work and payment of workers. Another potentially interesting application is in the eRFx procurement technology market. In procurement technology, eRFx is an acronym for Electronic Request For [x], where x can be Proposal (RFP), Quotation (RFQ), Information (RFI), or Tender (RFT). A huge amount of investment in the typical enterprise is communicated and tendered through this procurement process. By leveraging smart contracts and linking them to the new talent delivery model we can significantly streamline the entire process.

Open source projects in the smart contracts space

- Hyperledger Foundation (under the Linux Foundation)
- Accord Project
- Chainlink
- Waceo

As an application, the smart contract lends itself much more readily to the 'matching' process. It also provides invaluable information to 'shift-left' the visibility of demand skills to the talent community and vendors alike. Fulfilment and payment milestones could be automated because of smart contracts, to name just a few benefits.

Another entity associated with smart contracts is a decentralised autonomous organisation (DAO). A DAO is effectively a group of people who agree to abide by certain rules for a common purpose. Those rules are written into the code of the organisation via smart contracts on the blockchain (algorithms that run when certain criteria are met). Open source projects and their communities require a similar level of coordination and governance.

Given that the next iteration of the internet, web3, is built upon blockchain and is intrinsically linked to the metaverse, there is a potential opportunity to rethink where code is developed and versioned, as well as how to foster collaboration, provide training, and support communities to practice in the open source community. DAOs could be a vehicle to provide the governance layer for that vision.

Digital twins

For the metaverse to become a genuine place for work, it will be necessary to replicate assets ('assets' being anything from a container through to an entire operating system and everything in between). According to Accenture's *Technology Trends 2022* report, the global digital twin market was valued at US$3.21 billion in 2020 and is expected to grow to US$184.5 billion by 2030. IBM breaks digital twins into four categories:

- Component/Parts twins
- Asset twins
- System/Unit twins
- Process twins

Microsoft is the frontrunner in this exciting space with its Azure Digital Twins offering, but there are clear opportunities for the open source community to investigate the observability aspect of digital twin technologies.

Listed above are some of the critical components for the metaverse and its growth trajectory that open source is already involved in. There are clearly opportunities for open source oriented organisations to both engage with these open source projects and foundations. That engagement could take the form of thought leadership, evangelism, and training, as well as technical contributions to address some of the persisting challenges surrounding the metaverse such as privacy and security concerns.

However, there is a bigger opportunity that presents itself here. Namely, what could open source software development and communities look like in web3 and the metaverse? The modern technologies and approaches involved offer a unique opportunity to redesign it, to usher in an evolutionary step forward for open source. EFY

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